Spring 1996

Problem 1 (Sp96). Compute $L = \lim_{n\to\infty} \left(\dfrac{n^n}{n!}\right)^{1/n}$

Problem 2 (Sp86, Sp94, Sp96, Fa98). Let K be a compact subset of $\mathbb{R}^n$ and $\{B_j\}$ a sequence of open balls that covers K. Prove that there is a positive number ε such that each ε-ball centered at a point of K is contained in one of the balls B_j.

Problem 3 (Su77, Fa84, Sp94, Sp96, Sp09, Fa11). Evaluate the integral

$$I(a) = \int_0^{2\pi} \frac{\mathrm{d}\theta}{a + \cos\theta}$$

where a is real and $a > 1$. Why the formula obtained for $I(a)$ is also valid for certain complex (nonreal) values of a?

Problem 4 (Sp96). Let $r < 1 < R$. Show that for all sufficiently small $\varepsilon > 0$, the polynomial

$$p(z) = \varepsilon z^7 + z^2 + 1$$

has exactly five zeros (counted with their multiplicities) inside the annulus

$$r\varepsilon^{-1/5} < |z| < R\varepsilon^{-1/5}$$

Problem 5 (Sp96, Fa24). Prove or disprove: For any 2×2 matrix A over $\mathbb{C}$, there is a 2×2 matrix B such that $A = B^2$.

Problem 6 (Sp88, Sp96). If a finite homogeneous system of linear equations with rational coefficients has a nontrivial complex solution, need it have a nontrivial rational solution? Give a proof or a counterexample.

Problem 7 (Sp96). Prove that $f(x) = x^4 + x^3 + x^2 + 6x + 1$ is irreducible over $\mathbb{Q}$.

Problem 8 (Sp88, Sp96, Sp10). Determine the rightmost decimal digit of $A = 17^{17^{17}}$.

Problem 9 (Sp96). Exhibit infinitely many pairwise nonisomorphic quadratic extensions of $\mathbb{Q}$ and show they are pairwise nonisomorphic.

Problem 10 (Sp96). Show that a positive constant t can satisfy

$$e^x > x^t \qquad \text{for all } x > 0$$

if and only if $t < e$.

Problem 11 (Sp96). Suppose φ is a $\mathcal{C}^1$ function on $\mathbb{R}$ such that

$$\varphi(x) \to a \quad \text{and} \quad \varphi'(x) \to b \quad \text{as} \quad x \to \infty$$

Prove or give a counterexample: b must be zero.

Problem 12 (Sp96). Let $M_{2\times 2}$ be the space of 2×2 matrices over $\mathbb{R}$, identified in the usual way with $\mathbb{R}^4$. Let the function F from $M_{2\times 2}$ into $M_{2\times 2}$ be defined by

$$F(X) = X + X^2$$

Prove that the image of F contains a neighborhood of the origin.

Problem 13 (Sp96). Let $f = u + iv$ be analytic in a connected open set D, where u and v are real valued. Suppose there are real constants a, b and c such that $a^2 + b^2 \neq 0$ and

$$au + bv = c$$

in D. Show that f is constant in D.

Problem 14 (Su79, Sp82, Sp91, Sp96). Let f be a continuous complex valued function on $[0, 1]$, and define the function g by

$$g(z) = \int_0^1 f(t)e^{tz}\,\mathrm{d}t \qquad z \in \mathbb{C}$$

Prove that g is analytic in the entire complex plane.

Problem 15 (Fa80, Sp96). Suppose that A and B are real matrices such that $A^{\mathsf{T}} = A$,

$$v^{\mathsf{T}} A v \geqslant 0$$

for all $v \in \mathbb{R}^n$ and

$$AB + BA = 0$$

Show that $AB = BA = 0$ and give an example where neither A nor B is zero.

Problem 16 (Fa83, Sp96). Let A be the $n \times n$ matrix which has zeros on the main diagonal and ones everywhere else. Find the eigenvalues and eigenspaces of A and compute $\det A$.

Problem 17 (Sp96). Let G be the group of 2×2 matrices with determinant 1 over the four element field $\mathbb{F}$. Let S be the set of lines through the origin in $\mathbb{F}^2$. Show that G acts *faithfully* on S. An action is *faithful* if the only element of G which fixes every element of S is the identity.

Problem 18 (Fa87, Sp96). Let G and H be finite groups of relatively prime order. Show that $\mathrm{Aut}_{G \times H} \simeq \mathrm{Aut}_G \times \mathrm{Aut}_H$.